Xinyu (Rain) Wei

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Education

Columbia University | New York, NY Sep 2020 - May 2024

B.S. in Computer Science, Intelligent Systems Track, *summa cum laude* GPA: 4.092; top 5%

Honors and funding: Upsilon Pi Epsilon (2024); Bonomi Research Scholar (2023); Electrical Engineering Summer Research Grant (2023); Tau Beta Pi, junior inductee (2022); Mathematics Undergraduate Summer Research Fellowship (2022); Mathematical Modeling Research Fellowship (2021).

Teaching Assistant: Machine Learning (COMS 4771, graduate-level), Computer Science Theory (COMS 3261), and Discrete Mathematics (COMS 3203).

Coursework: 20 graduate and advanced technical courses, including Mathematics of Deep Learning, High-Dimensional Probability, Reliable Statistical Learning, Graphical Models, Robot Learning, and Convex Optimization.

Professional Experience

Capula Investment Management New York, NY & London, UK

Quantitative Analyst Oct 2024 - Present

- Devised an adaptive multi-task learning framework to improve prediction from sparse data across four bond-auction maturities. Used nonsquared-norm penalties to pool information while separating outliers, and a shared market-state factor with task-specific responses to model changing conditions. Integrated predictions with constrained trade timing and allocation, improving out-of-sample trading P&L by 20% relative to the desk's existing strategy.
- Formulated and deployed state-space and regression models to estimate changing fund sensitivities to market factors and identify pricing signals, using Kalman filtering, robust regression, and ridge regularization. Designed sequential allocation algorithms with isotonic projection and covariance-aware optimization to handle overlapping positions and correlated returns; automated hedge selection and sizing for live trading.
- Architected a research computation platform used by the entire team, separating interactive analysis from parallel batch execution. Designed shared caching with parameter-aware invalidation, resumable model fitting, and timeouts and failure isolation for approximately 20 computational jobs. Implemented process and thread pools for parallel workloads; tensorized and parallelized volatility-surface calculations, achieving a 5× computational speedup.

Quantitative Research and Trading Intern Jun 2024 - Sep 2024

- Formulated portfolio selection and allocation as a mixed-integer quadratic program, jointly choosing assets and position sizes under sparsity and concentration constraints. Implemented the optimization pipeline for a currency strategy deployed in live trading.

Research Experience

Columbia University New York, NY

Latent Express Traversal for Manifold Denoising Sep 2026 - Present

Faculty Advisors: Profs. John Wright and Szabolcs Márka

- Led development of the theoretical framework and independently designed the proof strategy for manifold denoising, linking geometric search to reconstruction accuracy through bounds on representation, noise, tangent-estimation, and discretization errors.
- Developed a more efficient graph-based denoising algorithm by consolidating redundant search connections, analyzing when a smaller search graph can preserve reconstruction accuracy.
- Extended experimental evaluation of the method as a geometric substitute for neural-autoencoder inference, examining reconstruction accuracy against memory usage and inference cost.

Trainable Template Optimization on Low-Dimensional Manifolds Mar 2023 - Jun 2024

Faculty Advisor: Prof. John Wright

- Led the core theoretical analysis of TpopT, which replaces exhaustive template-bank search with Riemannian optimization on a low-dimensional signal manifold. Independently designed the proof strategy, establishing curvature-dependent local linear convergence under Gaussian noise to a noise-dependent neighborhood. Proved accuracy bounds governed by intrinsic dimension with only logarithmic ambient-dimension dependence, and quantified the computational advantage over matched filtering as target accuracy increases.
- Research resulted in an arXiv preprint and a U.S. patent application.
- Invited presenter, NSF AI Institute in Dynamic Systems' Machine Learning for Science & Engineering Workshop (2023).

Research Experience (continued)

Neural Network Theory: Training Dynamics and the Neural Tangent Kernel Apr 2022 - Apr 2023
Faculty Advisor: Prof. John Wright

- Developed components of a time-dependent approximate-NTK framework for finite-width ReLU networks, deriving feature-perturbation bounds and analyzing activation-pattern changes to study training dynamics.
- Contributed proofs on local manifold geometry, smooth-function approximation, and small-norm kernel certificates in a related study of learning from noisy manifold observations.
- Implemented PyTorch experiments on synthetic circle and sinusoidal manifolds, tracking empirical NTKs across training checkpoints to examine within- and between-class kernel structure and directional responses to local input perturbations.

Adaptive Statistical Learning and Multi-Task Learning Aug 2022 - Jun 2024
Faculty Advisor: Prof. Kaizheng Wang

- Analyzed Gaussian multi-task estimation to characterize when pooling related tasks improves over separate estimation, deriving the bias–variance tradeoff and critical task-heterogeneity scale.
- Derived K-fold cross-validation estimators for regularized Gaussian mean estimation; evaluated statistical risk through Monte Carlo experiments and analytical comparisons with maximum-likelihood, Bayesian, and oracle benchmarks.

End-to-End Variational Inference for Portfolio Construction Apr 2023 - Jun 2024
Faculty Advisor: Prof. Garud Iyengar

- Derived task-loss gradients through constrained portfolio optimization using KKT-based implicit differentiation, supporting end-to-end learning of conditional return distributions.
- Contributed to the PyTorch implementation linking Monte Carlo samples from a variational generative model to differentiable portfolio optimization.
- Compared end-to-end learning with predict-then-optimize and direct allocation baselines in synthetic-data experiments, varying prediction and task-loss weights to study trade-offs between predictive accuracy and portfolio performance.
- Presented this work at Columbia DSI's Financial and Business Analytics Center (2023).

Publications & Patents

- J. Yan, S. Wang, **X. R. Wei**, Z. Márka, S. Márka, and J. Wright. *Systems and Methods for Efficient Trainable Template Optimization on Low Dimensional Manifolds for Use in Signal Detection.* U.S. Patent Application Publication US20250077872A1, 2025.
- J. Yan, S. Wang, **X. R. Wei**, J. Wang, Z. Márka, S. Márka, and J. Wright. *TpopT: Efficient Trainable Template Optimization on Low-Dimensional Manifolds.* arXiv:2310.10039, 2023.

Entrepreneurship & Performing Arts

Lokin | Co-founder Sep 2026 - Present

- Co-built and launched a trading research and backtesting platform with two co-founders; onboarded initial users.
- Led design of the agentic system and Research workspace, translating natural-language hypotheses into reviewable plans, executable analyses, and backtests; integrated research state, code validation and repair, and evidence-linked results.
- Designed Git-style strategy forking and version control, preserving immutable revisions and research lineage; enabled comparison of alternative strategies and linked results to versioned code, data, and execution settings.

International Latin Dance | Professional Dancer & Instructor 2004 - Present

- Professionally trained International Latin dancer competing in World Dance Council (WDC) and NDCA-recognized events; listed in NDCA rankings. Train approximately 15 hours weekly and provide private coaching and group lessons.
- Medalist and finalist at International Dance Festival, The Open Worlds, United States Dance Championship, Paragon Open Dancesport Championship, King's Ball, Philadelphia Dancesport Championship, Emerald Ball, and New York Dance Festival.
- 1st place, King's Ball Solo, Best of the Best category; lead dancer, Huayi Brothers Film Festival opening gala.

Elite Youth Education | Founder & Chief Executive Officer 2020 - 2024

- Founded and led an international education company, managing 70+ tutors and team members across curriculum development, operations, sales, and student delivery.
- Supported 200+ HSC and VCE students in achieving ATAR scores of 95+.

Technical Skills

Programming Languages:	Python, Java, C/C++, MATLAB, HTML/CSS, JavaScript, SQL, Ruby, Rust.
Scientific Computing:	PyTorch, JAX, NumPy, SciPy, scikit-learn, pandas, Matplotlib.
Developer Tools:	Unix/Linux, Jupyter Notebook, Conda, Shell (sh, bash, zsh), Git, GitHub, GitHub Actions, CI/CD, Docker, AWS, Microsoft Azure.